



Academic reactions to educational innovation

Peter B. Ayscough

To cite this article: Peter B. Ayscough (1976) Academic reactions to educational innovation, *Studies in Higher Education*, 1:1, 3-9, DOI: [10.1080/03075077612331376803](https://doi.org/10.1080/03075077612331376803)

To link to this article: <https://doi.org/10.1080/03075077612331376803>



Published online: 05 Aug 2006.



Submit your article to this journal [↗](#)



Article views: 93



View related articles [↗](#)



Citing articles: 1 View citing articles [↗](#)

Peter B. Ayscough *Professor in the Department of Physical Chemistry,
University of Leeds*

Academic Reactions to Educational Innovation

The path of the innovator is strewn with pitfalls, whatever the nature of his enterprise. One has no difficulty in choosing examples from such diverse fields as transport, medicine and politics, in which a promising innovation has gone seriously wrong or has been rejected for entirely unexpected reasons.

The educational innovator has perhaps a harder task than most since he necessarily operates in an environment with strong links with the past. After all, a teacher is obliged to relate the present state of his subject to its past history as well as its future development. The educational innovator is, more or less by definition, inviting a break with tradition and must therefore be viewed with suspicion by his colleagues. He is rarely able to state with any conviction that the application of his ideas is bound to be beneficial because he is generally not able to carry out the kind of experiment which might convince his audience. The problems of educational research are too well known to need stating here and it is only too easy for critics, especially those brought up in a rigorous scientific discipline, to dismiss the results on the grounds of lack of control, objectivity, statistical significance or whatever. Since the scientist does not willingly abandon one theory or practice for another without convincing evidence, it is hardly surprising that the pace of educational innovation is slow, particularly in science-based subjects. We are readily comforted by the thought that by and large our students do manage to acquire an education which provides them with useful skills and knowledge. Thus most of us manage to retain our composure while educational theorists and sociologists fight their battles in what seems an increasingly rarefied atmosphere not quite connected with our real world.

These thoughts have been generated as a result of personal experience in two spheres of educational innovation, one concerning educational technology, the other concerning curriculum development—though the two are not unconnected, as we shall see.

Computer Assisted Learning

Computer assisted learning (CAL) may, for the purpose of this paper, be classified as an application of educational technology, though its implications go much deeper and it is not possible to confine one's attention to the purely technical aspects for long. My own involvement started in an apparently innocuous manner, with a request for co-operation in defining a suitable model for an investigation of various strategies of CAL in a scientific area.

It so happened that I had recently taken over responsibility for running the Physical Chemistry teaching laboratories at Leeds, in the heady days when we were planning for ever-increasing numbers of Chemistry students. The problem facing us was that of providing the most valuable

educational experience for our students without greatly increasing the demands on staff time and laboratory space. Of the various solutions considered, the most promising appeared to be to define the objectives of the laboratory work more precisely and then rationalize the distribution of time in order to achieve these objectives economically. This resulted in an attempt to reduce the amount of repetitive experimental work (except where this was aimed at the development of a technical skill) and concentrate on the decision-making aspects of the course, i.e. in the choice of experimental methods appropriate to the chemical problems being studied.

The interrelated skills of decision-making and problem-solving are widely accepted as important, yet they are rarely given much priority in university curricula. One reason is that they are difficult to teach and require much expenditure of staff time. The experience should be individual, and, to be effective, opportunities should be provided to assess the consequences of decisions, whether correct or incorrect. We were able to consider adopting this approach as a result of the possibilities afforded by the kind of adaptive CAL system then being developed by the Leeds Computer Based Learning Project.

The series of investigations which resulted from the ensuing collaboration have been reported elsewhere (Ayscough, 1973; Abbatt & Hartley, 1974). They were carried out in the traditional style of educational research and led to conclusions about the value of CAL in this context which were significant, even encouraging. The one result which was totally unexpected was an invitation to direct a much larger development project in computer assisted learning, involving several other universities and polytechnics, based substantially on the kind of work pioneered in our own laboratories.

This project (CALCHEM) is hardly two years old, yet it can be said to have achieved many of its objectives already. CALCHEM (Computer Assisted Learning in Chemistry) is Development Project DP 1/06 of the National Development Programme in Computer Assisted Learning. In particular it has been demonstrated that this new educational technique is widely acceptable to students and their teachers if used with skill and discretion. This is not to say that all, or even the majority of chemistry teachers wish to embrace this innovation; what can be said is that computer assisted learning methods are being used in a dozen or more Chemistry departments throughout the country, and that CAL is being allowed to prove itself in an essentially friendly atmosphere.

It is too early to say that the future of CAL is secure, particularly in a period of financial stringency, but the history of CALCHEM will surely make an interesting case study, when it can be written. Meanwhile one hopes that one is not tempting providence by trying to draw some general conclusions concerning the extremely varied reactions of academic staff who have taken the trouble to find out what is going on. A place in the driving seat of a project with objectives of evaluation and dissemination puts one in a peculiarly favourable vantage point for observing those reactions.

Reactions to Computer Assisted Learning

In order to structure this essay let me make two assertions concerning educational innovation which may be construed more as articles of faith than as representations of reality. These are (1) good innovations rest on perceived needs, (2) such innovations will ultimately be accepted by other teachers who recognize similar needs in their own situations.

Both statements need to be qualified and expanded. By a 'good' innovation I mean one which is seen to be worthwhile in its immediate context and which will still be used even when the special situation which gave rise to it no longer applies. For example, the use of CAL in the teaching laboratories at Leeds arose as a response to a particular problem but continues now in its own right. In the second assertion the operative word is 'ultimately', because of the fundamentally conservative attitude of the average academic which means that any innovation will be subjected to a long period of critical study, as it should be, of course.

There are many obstacles to overcome before an innovation is accepted by an individual teacher and even more before it is accepted by a department or a profession. Without suggesting that the individual academic consciously adopts such a systematic approach, or even recognizes the existence of these obstacles, it seems that the six steps to salvation may be summarized as follows:

- (1) predisposition to consider innovation,
- (2) understanding what the innovation is and what it can do,
- (3) recognition that the innovation is relevant to one's own needs,
- (4) acceptance of evidence of value of the innovation,
- (5) willingness to learn how to use it,
- (6) identifying a potential personal contribution.

The head of a department or institution will have to consider additional factors such as cost (which does not normally concern the individual), effects on curricula, effects on other members of staff, relevance to objectives and attitudes at departmental level, etc. These matters are beyond the scope of this article.

Initiation

Obviously no progress at all can be made unless the teacher is willing to admit that some kind of educational innovation may be desirable. However, one must recognize that reluctance even to consider change is not necessarily evidence of a fundamentally Luddite attitude but may arise from quite different causes, including the precognition of some of the more distant consequences of displaying interest. For example, a colleague may be inhibited by the fact that sooner or later he may be required to examine his own activities in a critical manner and possibly to learn a new technique (step 5). Others may have learned from bitter experience how often an ingenuous proposal made at, say, a staff meeting, can lead to an invitation to follow it up oneself—plenty of encouraging noises but little actual help! Reassurance on this point may persuade the reluctant academic to proceed to the next stage at least.

The next three stages represent a logical approach to a new educational technique or procedure, but in practice the three hurdles are not necessarily approached in that order, nor need they be taken at one stride. For most of us the knowledge of what the innovation can do is initially fragmentary and our consideration of its relevance to us is casual. First impressions are clearly of great importance, for we are unlikely to be motivated to proceed further unless these are favourable and we can already see some relevance to our own situation.

Using the example of CAL again, one suspects that the conventional image of CAL for many people is that of the large American freshman class performing some trivial task such as a simulated titration by means of a computer. This is sufficient to cause total rejection by those who cannot foresee needs of this kind in their own departments and would be horrified if they did. For others the introduction may have been by means of a sophisticated demonstration of three-dimensional molecular structures on a television screen—very clever but surely outrageously expensive. The best demonstration of all will almost certainly involve watching students at work in an environment similar to one's own.

Given this kind of introduction the well-disposed enquirer will already have begun to see possibilities which he may then relate to his own needs. Such a person will by now have an incentive to learn more about what the innovation can do and in particular to look more closely at the evidence that it works in the way it is intended to. We will look at the approach to this evidence in a moment. Meanwhile we must consider the person who has been prejudiced by his early experience. It may be possible to counteract this prejudice by providing more acceptable examples but the effort will be greater and the evidence will have to be even more convincing.

In CALCHEM we have encountered several situations of this kind (both individual and

departmental) and have no doubts of the extent of the prejudice which can arise from early impressions caused by faulty or inappropriate demonstrations, incorrect information or unrealistic claims. On the other hand, where the system has survived the more critical examination to which it has been subjected after such an inauspicious start, the early prejudice can be set aside and a convert gained—and we know how often converts become more zealous than those brought up in the faith. The other side of the coin is exemplified by the enthusiast whose acceptance of the technique may be based on inadequate knowledge of its limitations and can therefore become embarrassing at a later stage.

Evaluation

It seems impossible in the world of educational innovation to achieve a truly objective assessment of the evidence because this is itself generally so complex and so heavily qualified. Also, as in the case of highly specialized scientific studies, the vocabulary (jargon?) can render the evidence unintelligible to the uninitiated and once again the element of prejudice may become evident. In CALCHEM we have tried to avoid both the pseudo-objectivity of the conventional control group/pre- and post-test analysis, and the temptation to wrap our observations in the verbal trappings of educational research (Ayscough, 1976). We are, after all, trying to address ourselves to practising teachers who are not particularly interested in educational research as such. We may not succeed in doing so indefinitely but, so far, the project, which started from a very small nucleus, has expanded very satisfactorily without the support of weighty analysis and impressive publications. Our conclusion is that *those who have pursued their investigations of CAL to the point at which they have a reasonable grasp of what we are doing, recognize that the real evidence they seek can only be obtained by observing the system in action*, and by assessing the reactions of its users. However, one also realizes that many of the most searching questions relate to the academic respectability of those concerned with designing and operating the system (there is a moral here which perhaps need not be spelled out!). Consequently dissemination of this particular innovation has occurred almost entirely on a person-to-person basis and by direct observation, *not* by evangelical contributions to seminars and journals.

It is worth considering, again in relation to CALCHEM, the cases where an enquirer has failed to get past one or other of these three hurdles. In our rather limited experience so far the most common reasons seem to be (1) failure of the CAL system to live up to expectations (not the expectations generated by our own claims, but those arising from lack of knowledge and/or realism), (2) failure to accept the relevance of CAL to their own situation (in many cases this means disagreeing with the essential objectives of the innovation). Under neither of these circumstances is further argument or persuasion likely to be fruitful.

Consummation

Given at least qualified acceptance of the nature of the innovation, its validity, purpose and relevance, the innovator must now take account of the problems involved in persuading the teacher to become a learner and then providing him with an opportunity to make his own individual contribution (items 5 and 6). Each is potentially crucial to success.

The reluctance of most academics to learn new educational methods is very well known. One needs only to look at the attendance at seminars and courses on teaching techniques in one's own university to recognize this, and the difficulty many of us have in putting ourselves in the role of learner, particularly in public, cannot be overstated. The emotive aspects of the problem are readily apparent, yet they must be overcome if the innovation is to be implemented successfully. Some of the most striking differences in attitudes observed during the formative stages of the CALCHEM project relate to this question and we have to admit to some notable failures which are perhaps better not subjected to too much public scrutiny. Incidentally, we have found that

the workshop format is most successful in creating a favourable atmosphere for learning, particularly if the most experienced participants also place themselves in the position of learner by soliciting assistance in the solution of their own problems. If this sounds like group therapy perhaps we have to admit that there is something to be learned from psychology!

Curiously, and very fortunately, we do not have this reluctance to learn new tricks in relation to our research activities. Since human beings are from birth essentially explorers and innovators, one must suppose that in academics this dichotomy arises from the relative rewards of the two activities and the competitive career structure within which we operate. Advantage can be taken of this situation if the introduction and development of the innovation can be seen as a research activity, with the consequential benefit of publications, etc. This incentive was denied to most of those involved in CALCHEM in the early stages by the project's sponsors but has now been at least partially restored.

Although not strictly relevant to the argument being presented, one can see that, of the many roles which an academic will accept under duress, that of guinea-pig is particularly distasteful. Furthermore, many of us are reluctant to place our students in this role and one wonders how much educational research has foundered or been substantially distorted by this factor. Certainly CALCHEM would never have survived if this aspect had not been handled with great tact by those concerned with evaluation, and accepted with understanding by those collaborating in the project.

Returning to the main theme, the desire to seize on some aspect of the innovation and use it as a vehicle for one's own imagination is probably the most powerful incentive of all for the potential innovator, and one can predict with some certainty that the innovation will not be accepted with any enthusiasm by any teacher unless this opportunity is provided. It is very fortunate that the ethos, as well as the methodology, of CAL is still in a formative stage, despite a decade or more of development. It is thus possible to attract and retain the interest of imaginative and energetic academics who would otherwise find some other outlet for their energy.

I believe that this is the key to the present success of CALCHEM and may have some wider significance. It is necessary to explain that CALCHEM is in effect a loose federation in which members of a dozen or so departments collaborate in the design, production and testing of CAL programs. A large proportion of the actual programming and pre-testing is done centrally (at Leeds University and Sheffield Polytechnic) but the trials are carried out independently in each of the collaborating institutions—with collation and administrative assistance from the project base in Leeds. From the outset it was recognized that the educational and financial evaluation of CAL, which is a major objective of the project, would not provide an adequate incentive to the teachers involved: in fact it is viewed by most as a necessary but unwelcome burden. The academics concerned are lecturers in departments of Chemistry, not of Education, so they are not impelled by any desire to publish the results of exercises in evaluation. However, they are much more likely to co-operate in achieving the project's overall objectives if they can see a distinctive role for themselves. This has meant accepting, and even encouraging, ventures into new applications alongside the production of more standard materials. This is administratively untidy, it costs more so our sponsors are not enthusiastic, but it is essential for the health of the project. The new products may not always be better than the old, but they will have served their purpose if they have helped to maintain the enthusiasm of their creators while permitting the continuation of the more humdrum development and evaluation exercises.

There is another level at which the necessity of putting one's own individual stamp on the product is essential and is being catered for. That is the point at which a CAL program is transferred from one department to another. (Such transfers are essential in CAL in order to spread the development costs over a larger number of users and to provide a variety of environments in which to study student attitudes and performance.) Since most of our programs are in the nature of a tutorial treatment of an experimental or theoretical problem, they inevitably reflect the style, attitude, phraseology and terminology of their creators. There may also be detailed

differences in the way a particular problem is approached by two different teachers and it must be possible to take this into account. The fact that our programs can be readily modified to suit local needs is a very important factor in bringing about their acceptance. (One must also point out the converse: where local modifications cannot be made, as in the case of pre-packed educational materials such as film, TV and programmed text, the imported material may be rejected on the basis of an essentially trivial objection concerning, for example, terminology. This is, I believe, one of the major obstacles to the widespread transfer of what may otherwise be excellent educational materials.)

I have devoted considerable space to matters which are simply concessions to academic individuality and independence, but these aspects are ignored by innovators at their peril. The problem arises at departmental or institutional level as well, particularly where the innovation is closely associated with one specific institution or group. The importance of flexibility of approach which permits the receiving department to adopt its own style of implementation (even when one is convinced that this is not the best) cannot be over-stated. It may appear to be wasteful of resources but it is the price one must be prepared to pay for wholehearted co-operation rather than grudging acceptance. As in other aspects of education, learning by discovery, even at some cost, may be the only way to achieve the necessary motivation for continued study.

Chemical Sciences: Innovation in Curriculum

The introduction of a totally new and additional degree scheme by the School of Chemistry at Leeds is a major innovation of a different kind and is fraught with many of the same problems as computer assisted learning. Because of its emphasis on the unified structure of the subject and the breadth of its applications, it demands a much higher level of co-operation between academic staff than is usual, and also a willingness to consider a radically different approach, including a rather detailed planning of the whole syllabus. The introduction of this course has raised contentious issues in much the same way as computer assisted learning and it is interesting that the checklist of stages through which an innovation must pass before winning approval appears to have some validity in this context as well as that of CAL. However, the discussion must be more circumspect and any conclusions even more tentative.

It is no secret that at the moment the attitudes of the staff of the School of Chemistry cover the whole spectrum from enthusiastic and supportive to indifferent and even hostile. Whether or not these attitudes have been arrived at by any process resembling the six steps outlined earlier cannot be established, but one can identify among one's colleagues a number whose disapproval may be correlated with their reluctance to proceed beyond one or other of those stages.

For example, there is a minority who resist any form of change and one imagines that they have their counterparts in every educational establishment. There are those who genuinely do not accept the validity of the alternative approach to Chemistry exemplified by this course, or, equally sincerely, see no need for it. Then there are some who accept the validity of the course and its relevance but do not consider it appropriate for a university department which already runs an excellent conventional course: they argue that the stretching of resources to cover two degree courses is to the ultimate advantage of neither. Others who accept all the arguments in favour of the new course are still far from enthusiastic about participating: one suspects that they are in the category of reluctant learners. They realize that they may have to lecture on unfamiliar topics or adopt unfamiliar approaches and believe that they may not find this agreeable. Others go the whole way in supporting the innovation and then proceed to adapt it in some way to their own viewpoint, which may not be entirely compatible with the objectives of the course. However they would not be good academics if they did otherwise and, as in the case of CALCHEM described earlier, the price of their wholehearted co-operation is some concession to their individuality.

Conclusions

This brief essay is an attempt to make a number of generalizations about reactions of academic staff to two kinds of educational innovation. Since it is based on personal observation of a limited sample any conclusions must be tentative and to some extent superficial. The approach has been to try to relate these reactions to a logical sequence of stages presented as barriers to further progress. Provided that one accepts that many academics are sufficiently perceptive to realize the implications of some of the later stages and therefore decide to opt out earlier, there appears to be some support for this kind of analysis. This being so, the lesson for the innovator is that he must devise a strategy which takes account of the many different stages at which teachers will resist the innovation and then take steps to accommodate their justifiable needs and criticisms.

While the logic of this approach may be accepted, one must be aware of its limitations. It takes no account of the psychological factors affecting motivation or response, nor of the relationships between personalities; it assumes that the academic is prepared to be objective most of the way. (In the last resort he will not, since his decision whether or not to become involved will include many subjective issues which cannot be taken into account in this kind of analysis.) But I believe that, by and large, these additional factors can be treated as variables which cause relatively minor perturbations of the scheme proposed.

Finally, to those of my colleagues and associates who see reflections of themselves in the people portrayed in this article, I offer my apologies. By stressing the problems of innovation I have inevitably presented a one-sided account; my predominant reaction to the challenge of introducing innovation has been that of pleasure of working alongside colleagues with so much tolerance, understanding and personal warmth. My sincere thanks to them all.

Correspondence: Dr P.B. Ayscough, Department of Physical Chemistry, University of Leeds, Leeds LS2 9JT.

REFERENCES

- ABBATT, F.R. & HARTLEY, J.R. (1974) Teaching planning skills by computer, *Journal of Mathematical Education in Science and Technology*, 5, pp. 665–671.
- AYSCOUGH, P.B. (1973) Computer-based learning in the teaching laboratory, *Chemistry in Britain*, 9, pp. 61–65.
- AYSCOUGH, P.B. (1976) CALCHEM—An Exercise in Evaluation, *Computers in Education*, 1.